AP BIOLOGY TOPIC 2.10

Origins of Cell Compartmentalization: Endosymbiotic Theory & Evolution

UNIT 2: CELL STRUCTURE AND FUNCTION (10–13% EXAM WEIGHT)

ENE-2.M | **EVO-1.A** | **ENDOSYMBIOTIC THEORY** | **MEMBRANE INFOLDING**

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 2.10 (ORIGINS OF CELL COMPARTMENTALIZATION: ENDOSYMBIOTIC THEORY & EVOLUTION)

CED Core Vocabulary

Mitochondria and chloroplasts — evolved from once free-living prokaryotes (bacteria) that were engulfed via phagocytosis by an ancestral host cell (an anaerobic archaeon/proto-eukaryote),...

Evolutionary Significance of Endosymbiosis — Mitochondria provided massive aerobic ATP production (~30–32 ATP/glucose vs. 2 ATP from glycolysis) across vast internal cristae membranes.

Result — Enabled the evolution of much larger, structurally complex eukaryotic cells and eventually multicellular organisms with specialized tissues.

2. Carboxysomes (Bacterial Microcompartments) — Polyhedral protein shells (NOT lipid membranes!) found in cyanobacteria and chemoautotrophs that encapsulate RuBisCO and carbonic anhydrase → concentrate CO_2...

Nearly ALL eukaryotes — (animals, plants, fungi, protists) have mitochondria (or mitochondrial remnants), whereas only photosynthetic eukaryotes (plants & algae) have chloroplasts .

Chloroplasts — gave eukaryotic algae & plants autotrophic carbon fixation via oxygenic photosynthesis.

1. The Nucleoid Region — Non-membrane-bound region where the circular bacterial chromosome is condensed and organized by nucleoid-associated proteins.

§2. TWO EVOLUTIONARY ORIGINS OF EUKARYOTIC ORGANELLES

Infolding vs. Endosymbiosis

- **1. Origin of the Endomembrane System (Autogenous Membrane Infolding):**
  - The **nuclear envelope, endoplasmic reticulum (ER), Golgi apparatus, lysosomes, and vacuoles** evolved from **inward foldings (invaginations)** of the ancestral archaeal host cell's own plasma membrane (~2.1 billion years ago).
  - Explains why endomembrane organelles have **single lipid bilayers** (or a contiguous double fold in the nuclear envelope) and why their internal lumen is topologically equivalent to the exterior of the cell.
- **2. Origin of Mitochondria & Chloroplasts (Endosymbiotic Theory — Lynn Margulis):**
  - **Mitochondria and chloroplasts** evolved from once **free-living prokaryotes (bacteria)** that were engulfed via phagocytosis by an ancestral host cell (an anaerobic archaeon/proto-eukaryote), escaped digestion, and established a mutually beneficial **endosymbiosis**.
  - Over evolutionary time, extensive horizontal gene transfer moved many endosymbiont genes into the host nucleus, making the host and organelle **mutually interdependent (1 single organism)**.

1. Membrane Infolding → Nucleus & ER →

2. Engulf Aerobic α-Proteobacterium → Mitochondria (ALL Eukaryotes!) →

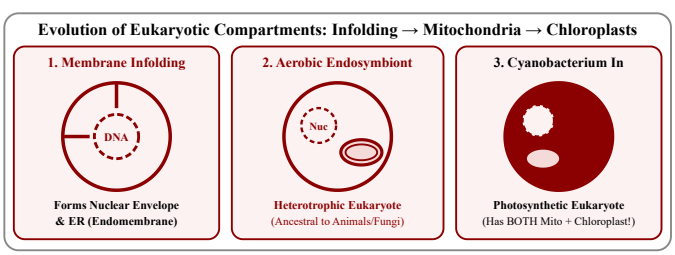
3. Engulf Cyanobacterium → Chloroplasts (Plants & Algae)

[AP EXAM FRQ SCORING STRATEGY] **Serial Endosymbiosis Order (Why Mitochondria Came FIRST!):**

- **Nearly ALL eukaryotes** (animals, plants, fungi, protists) have **mitochondria** (or mitochondrial remnants), whereas **only photosynthetic eukaryotes** (plants & algae) have **chloroplasts**.
- Therefore, the engulfment of the **aerobic prokaryote (mitochondrion) occurred FIRST** in the common ancestor of all eukaryotes; later, one lineage of mitochondrion-bearing eukaryotes engulfed a **photosynthetic cyanobacterium (chloroplast)**!

§3. MEMBRANE INFOLDING & SERIAL ENDOSYMBIOSIS VISUALIZED

Evolutionary Timeline



§4. 5 KEY LINES OF EVIDENCE FOR ENDOSYMBIOSIS

High-Yield FRQ List

Mitochondria and chloroplasts share five diagnostic structural, genetic, and biochemical traits with modern **prokaryotes (bacteria)**:

LINE OF EVIDENCE	IN MITOCHONDRIA & CHLOROPLASTS	WHY IT SUPPORTS BACTERIAL ORIGIN
1. Double Membrane	Surrounded by **two phospholipid bilayers** (inner + outer membrane)	**Inner membrane** = original bacterial plasma membrane (has bacterial cardiolipin & ETC); **Outer membrane** = derived from host's phagocytic vesicle!
2. Own Circular, Naked DNA	Contain their own **single, circular DNA molecule** lacking histone proteins	Matches **bacterial chromosomes/plasmids** (unlike linear, histone-wrapped nuclear eukaryotic chromosomes)
3. Own 70S Ribosomes	Transcribe & translate their own proteins using internal **70S ribosomes**	Prokaryotes have **70S ribosomes** (sensitive to bacterial antibiotics like streptomycin/tetracycline), whereas eukaryotic cytosol has **80S ribosomes**!
4. Independent Binary Fission	Replicate inside the cell by **splitting in two (binary fission)** independently of mitosis	Same asexual division mechanism (using FtsZ proteins) as free-living bacteria; if all mitochondria are removed, a cell cannot make new ones from scratch!
5. Size, Porins & Phylogeny	~1–5 μm size; outer membrane has bacterial **porin** proteins; rRNA sequences	Mitochondrial rRNA groups phylogenetically with **α-proteobacteria**; chloroplast rRNA groups with **cyanobacteria**

[AP EXAM FRQ SCORING STRATEGY] **FRQ Scoring Rule:** Always state **at least 3 distinct pieces of evidence** when asked to justify Endosymbiotic Theory: (1) **Double membrane**, (2) **Own circular naked DNA**, (3) **Own 70S ribosomes**, and (4) **Reproduce via binary fission**!

[COMMON AP EXAM FRQ TRAP] **Common Ribosome Mix-Up:** Never confuse **70S ribosomes** (found in *prokaryotes, mitochondria, and chloroplasts*) with **80S ribosomes** (found in the *eukaryotic cytosol and on Rough ER*)!

§5. EVOLUTIONARY SIGNIFICANCE & PROKARYOTIC MICRODOMAINS

Prokaryotes vs. Eukaryotes

- **Evolutionary Significance of Endosymbiosis:**
  - **Mitochondria** provided massive aerobic ATP production (~30–32 ATP/glucose vs. 2 ATP from glycolysis) across vast internal cristae membranes.
  - **Chloroplasts** gave eukaryotic algae & plants autotrophic carbon fixation via oxygenic photosynthesis.
  - **Result:** Enabled the evolution of **much larger, structurally complex eukaryotic cells** and eventually **multicellular organisms** with specialized tissues.
- **Do Prokaryotes Have Compartmentalization? (Prokaryotic Microdomains):**
  - Prokaryotes (Bacteria & Archaea) **lack internal membrane-bound organelles**, but they still organize metabolic pathways into specialized **non-membrane regions / protein microcompartments**:
  - **1. The Nucleoid Region:** Non-membrane-bound region where the circular bacterial chromosome is condensed and organized by nucleoid-associated proteins.
  - **2. Carboxysomes (Bacterial Microcompartments):** Polyhedral **protein shells** (NOT lipid membranes!) found in **cyanobacteria** and chemoautotrophs that encapsulate **RuBisCO** and carbonic anhydrase → concentrate **CO_2** around RuBisCO to boost Calvin cycle carbon fixation efficiency and block photorespiration!
  - **3. Infalded Plasma Membrane Regions (Thylakoid-like / Mesosomes):** Cyanobacteria fold their own plasma membrane inward to house photosynthetic electron transport chains.

[CORE AXIOM / REMEMBER] **Prokaryotes = Open-Plan Workspace with Organized Zones:** No internal lipid walls (no membrane-bound organelles), but organized functional zones like the **nucleoid** and protein-shelled **carboxysomes**!

[COMMON AP EXAM FRQ TRAP] **Nucleoid ≠ Nucleus!** A **nucleus** is a double-membrane-bound eukaryotic organelle; a **nucleoid** is simply the non-membrane-bound cytoplasmic region where prokaryotic circular DNA resides.

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§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 2.10 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 2.10: Origins of Cell Compartmentalization: Endosymbiotic Theory & Evolution ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 4 (Diffusion & Osmosis): Dialysis tubing & potato core mass change (% ΔMass = [(Final − Initial) / Initial] × 100) across sucrose molarities (0.0 M to 1.0 M). IV: Extracellular sucrose molarity (M); DV: % change in mass; Isotonic point: x-intercept where % ΔMass = 0% (no net water movement). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Graph Rules: Total Water Potential: $\Psi = \Psi_p + \Psi_s$. Solute Potential: $\Psi_s = -iCRT$ (where i = ionization constant [1 for sucrose/glucose, 2 for NaCl], C = molarity in mol/L, R = 0.0831 L·bar/mol·K, T = K = °C + 273). Surface-Area-to-Volume Ratio (SA:V): Cube SA = $6s^2$, V = s^3 (SA:V = 6/s). Larger SA:V accelerates diffusion per unit volume. **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Physiological Disruption Case Study: Cystic Fibrosis results from a deletion (ΔF508) in the CFTR chloride channel protein, preventing misfolded channels from reaching the apical plasma membrane. Impaired Cl^- efflux prevents osmotic water movement into mucus, producing thick, dehydrated mucus in lungs and pancreatic ducts. Ouabain/Digoxin inhibits the Na^+/K^+-ATPase pump, collapsing the electrochemical gradient required for secondary active cotransport.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 2.10 to prove why the claim holds.

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